



Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.

UNIT 8 • STANDARD 6.SP.4

Display Data: Box Plots

Lesson 8-5 • Teacher Copy

Name: _____

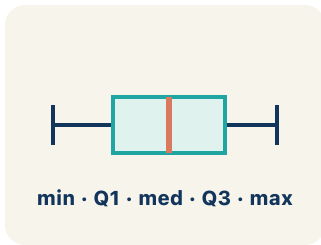
Date: _____

Class: _____

Key Vocabulary

Level 2 Standard

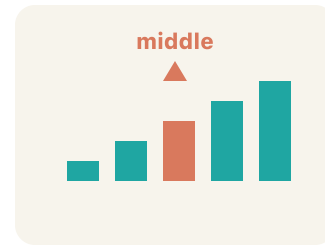
Picture first, then the word, then a plain-language meaning. Say each word out loud.



A box from Q1 to Q3 with a line at the median, and whiskers from min to max

Box Plot

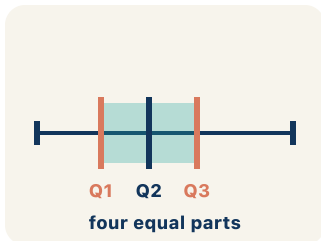
Write the definition:



Data: 10, 15, 20, 25, 30 → median is 20 (the 3rd value)

Median

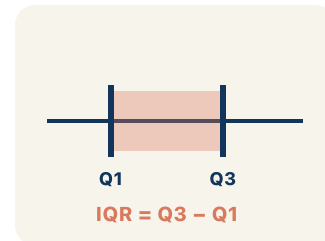
Write the definition:



Data: 2, 4, 6, 8, 10, 12, 14 → Q1 = 4 (median of lower half), Q3 = 12 (median of upper half)

Quartile

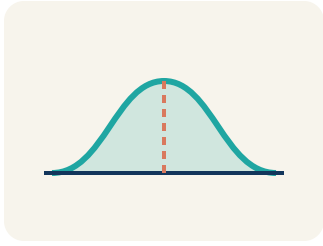
Write the definition:



If Q1 = 12 and Q3 = 20, then IQR = 20 - 12 = 8 — the middle 50% spans 8 units

Interquartile Range

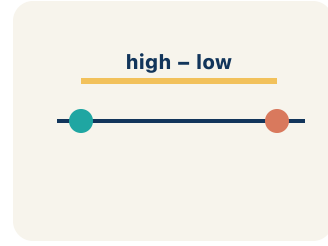
Write the definition:



A box plot where the median is centered in the box shows symmetric distribution

Data distribution

Write the definition:



*Small IQR = low variability in the middle 50%.
Large IQR = high variability*

Variability

Write the definition:

Guided Notes

Level 2 Standard



WHAT WE'RE LEARNING TODAY

I can make and read a box plot to summarize a data set.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Box Plot

Median

Quartile

Interquartile Range

Data distribution

Variability



Tap any word to see what it means and a picture.

1 A graph that shows data with a box and whiskers based on the five-number summary is a .

2 The middle value of the data, shown as a line inside the box, is the .

3 A value that divides the data into four equal parts is a .

4 The distance between the first and third quartiles, showing the middle 50% of data, is the .

5 The way data values are spread out is the .

6 How spread out the data values are is the .



Watch & Try — Worked Examples

See the notes in action: watch one worked all the way through, then try the next with the same steps.

 I do — watch

Follow each step as your teacher solves it.

Problem: A box plot shows: Min = 10, Q1 = 15, Median = 20, Q3 = 28, Max = 35. What is the interquartile range (IQR)?

- A. 13
- B. 25
- C. 10
- D. 20

Step 1 IQR = $Q3 - Q1 = 28 - 15 = 13$.

 **Answer:** A. 13

 We do — together

Solve this one with your class using the same steps.

Problem: On a box plot, what does the line inside the box represent?

- A. The median
- B. The mean
- C. The mode
- D. The range

Step 1 _____

Step 2 _____

Answer: _____

 You do — your turn

Now try one on your own. Show every step.

Problem: What percentage of data falls between Q1 and Q3 on a box plot?

- A. 50%
- B. 25%
- C. 75%
- D. 100%

Show your work:

Try It

Solve on your own. Check the answer key when you are done.

1. The Comets scored 6, 10, 14, 18, 22, 26, 30, 34 across 8 games. After ordering, what are Q1 and Q3?

- A. $Q1 = 12$, $Q3 = 28$
- B. $Q1 = 10$, $Q3 = 30$
- C. $Q1 = 14$, $Q3 = 26$
- D. $Q1 = 20$, $Q3 = 34$

Show your work:

2. A box plot for the Rockets shows Min = 8, $Q1 = 15$, Median = 22, $Q3 = 30$, Max = 38. The middle 50% of their scores falls between which two values?

- A. Between 15 and 30
- B. Between 8 and 38
- C. Between 8 and 22
- D. Between 22 and 38

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

Two basketball teams played 9 games each. Team A: 40, 42, 44, 46, 48, 50, 52, 54, 56. Team B: 30, 35, 40, 45, 48, 50, 55, 60, 65. Find the five-number summary and IQR for each team. Which team is more consistent? Which has a wider spread in the middle 50%?

Sentence starter: Team A: Min=____, Q1=____, Median=____, Q3=____, Max=____, IQR=____. Team B: Min=____, Q1=____, Median=____, Q3=____, Max=____, IQR=____. Team ____ is more consistent because its IQR is ____.

Show your work:

Reflect — Exit Ticket

A box plot has $Q1 = 20$ and $Q3 = 36$. What is the IQR and what does it represent?

- A. IQR = 16; it is the spread of the middle 50% of the data
- B. IQR = 56; it is the total of Q1 and Q3
- C. IQR = 28; it is the median between Q1 and Q3
- D. IQR = 36; it is the value of Q3

Your answer:

Answer Key & Teacher Guide

1. **Notes 1:** Box Plot
2. **Notes 2:** Median
3. **Notes 3:** Quartile
4. **Notes 4:** Interquartile Range
5. **Notes 5:** Data distribution
6. **Notes 6:** Variability
7. **We Try (notes):** A. The median — *The line inside the box of a box plot always represents the median.*
8. **You Try (notes):** A. 50% — *The box in a box plot contains the middle 50% of the data (from Q1 to Q3).*
9. **Try It 1:** A. $Q1 = 12$, $Q3 = 28$ — *With 8 values the median splits the data into a lower half (6, 10, 14, 18) and an upper half (22, 26, 30, 34). Q1 is the median of the lower half: $(10 + 14) \div 2 = 12$. Q3 is the median of the upper half: $(26 + 30) \div 2 = 28$.*
10. **Try It 2:** A. Between 15 and 30 — *The box of a box plot stretches from Q1 to Q3, and that box always holds the middle 50% of the data. Here the box goes from $Q1 = 15$ to $Q3 = 30$.*
11. **Exit Ticket:** A. $IQR = 16$; it is the spread of the middle 50% of the data — $IQR = Q3 - Q1 = 36 - 20 = 16$. *The IQR tells you the range of the middle 50% of the data.*